Big Data Practice Questions

89 MCQs with answers

HDFS and MapReduce

1. Which HDFS component is primarily responsible for monitoring replica counts and initiating re-replication when a block becomes under-replicated?

A. DataNode
B. Client
C. NameNode
D. Secondary NameNode

Answer: C. NameNode

2. What is the primary purpose of introducing the HDFS Federation?

A. Increase replication factor
B. Increase namespace scalability
C. Reduce block size
D. Eliminate DataNodes

Answer: B. Increase namespace scalability

3. Which of the following is directly affected by the number of HDFS blocks?

A. NameNode memory usage
B. Number of Map tasks
C. Both A and B
D. Neither A nor B

Answer: C. Both A and B

4. What is the default value of the parameter dfs.replication?

A. 1
B. 2
C. 3
D. 4

Answer: C. 3

5. If a DataNode stops sending heartbeats, what does the NameNode do?

A. Ignores the failure
B. Deletes all blocks immediately
C. Marks the DataNode and initiates re-replication if needed
D. Restarts the DataNode automatically

Answer: C. Marks the DataNode and initiates re-replication if needed

6. Which statement about Reduce tasks is TRUE?

A. A key can be processed by multiple Reducers.
B. Each key is assigned to exactly one Reducer.
C. Reducers execute before Mappers.
D. Reducers never use partitioning.

Answer: B. Each key is assigned to exactly one Reducer.

7. What is the primary role of a Partitioner in MapReduce?

A. Sort records within a Reducer
B. Decide which Reducer receives a key-value pair
C. Combine Mapper outputs locally
D. Replicate data blocks across HDFS

Answer: B. Decide which Reducer receives a key-value pair

8. Which YARN component is responsible for detecting task failures for a specific job?

A. Resource Manager (RM)
B. Node Manager (NM)
C. Application Master (AM)
D. DataNode

Answer: C. Application Master (AM)

9. Which statement about the Reduce function is TRUE?

A. It accepts an intermediate key and a set of values.
B. It processes only one value at a time.
C. It cannot aggregate values.
D. It always produces multiple output values.

Answer: A. It accepts an intermediate key and a set of values.

10. In the Reverse Web-Link Graph application, the Mapper emits:

A. (source, target)
B. (target, source)
C. (URL, 1)
D. (target, count)

Answer: B. (target, source)

Hadoop Ecosystem and the Vs of Big Data

11. A dataset was highly useful in 2024, but by 2026 the same dataset becomes irrelevant because the information has lost significance. Which "V" best describes this situation?

A. Validity
B. Veracity
C. Volatility
D. Venue

Answer: C. Volatility

12. Which of the following statements about Oozie is CORRECT?

A. Oozie is a distributed NoSQL database.
B. Oozie is a workflow scheduler system for Hadoop jobs.
C. Oozie is used to store data in HDFS.
D. Oozie is a machine learning library for Spark.

Answer: B. Oozie is a workflow scheduler system for Hadoop jobs.

13. Oozie can directly coordinate which of the following Hadoop ecosystem components?

A. Hive, Pig, Sqoop, and MapReduce
B. HDFS, NameNode, DataNode, and YARN only
C. Cassandra, MongoDB, and Redis only
D. Spark MLlib and GraphX only

Answer: A. Hive, Pig, Sqoop, and MapReduce

14. What is the primary purpose of Apache Flume in the Hadoop ecosystem?

A. Querying data stored in HDFS
B. Collecting and transferring large amounts of log data
C. Performing machine learning tasks
D. Managing cluster resources

Answer: B. Collecting and transferring large amounts of log data

15. What is the primary role of Apache ZooKeeper in a Hadoop ecosystem?

A. Distributed data storage
B. Workflow scheduling
C. Distributed coordination and synchronization
D. SQL query processing

Answer: C. Distributed coordination and synchronization

16. Hive primarily operates on data stored in:

A. MongoDB
B. Cassandra
C. HDFS
D. Redis

Answer: C. HDFS

17. Which of the following best describes Apache Hive?

A. A distributed coordination service
B. A workflow scheduler
C. A data warehouse system built on Hadoop
D. A NoSQL database

Answer: C. A data warehouse system built on Hadoop

18. Which Hadoop module stores metadata?

A. DataNode
B. TaskTracker
C. NameNode
D. Reducer

Answer: C. NameNode

19. Which pair is correctly matched?

A. Hive - Stream Processing
B. Kafka - Distributed Commit Log
C. Sqoop - Graph Analytics
D. GraphX - ETL Tool

Answer: B. Kafka - Distributed Commit Log

Big Data Fundamentals

20. What is Big Data?

A. Small amount of structured data only
B. Data that is too large and complex for traditional processing systems
C. Data stored only in databases
D. Data generated only from social media

Answer: B. Data that is too large and complex for traditional processing systems

21. Which of the following is NOT one of the 5 Vs of Big Data?

A. Volume
B. Velocity
C. Variety
D. Validation

Answer: D. Validation

22. Which framework is widely used for distributed processing of Big Data?

A. Hadoop
B. MS Word
C. Notepad
D. Photoshop

Answer: A. Hadoop

23. What is the primary function of HDFS in Hadoop?

A. Data Visualization
B. Distributed Storage
C. Query Optimization
D. Machine Learning

Answer: B. Distributed Storage

24. In Hadoop, MapReduce consists of which two phases?

A. Read and Write
B. Input and Output
C. Map and Reduce
D. Store and Retrieve

Answer: C. Map and Reduce

25. Which database type is commonly used in Big Data environments?

A. NoSQL Database
B. Spreadsheet
C. Text File Only
D. Compiler

Answer: A. NoSQL Database

26. Apache Spark is faster than Hadoop MapReduce mainly because:

A. It uses more hardware
B. It processes data in memory
C. It stores less data
D. It does not use clusters

Answer: B. It processes data in memory

27. Which of the following is an example of unstructured data?

A. Excel Table
B. Relational Database Table
C. Video File
D. Student Mark Sheet

Answer: C. Video File

28. Which component of Hadoop manages resources and job scheduling?

A. HDFS
B. YARN
C. Hive
D. Pig

Answer: B. YARN

29. Which language is used by Apache Hive for querying data?

A. SQL
B. HiveQL
C. JavaScript
D. C++

Answer: B. HiveQL

Cassandra and ZooKeeper

30. Which Cassandra replication strategy is specifically designed for multi-datacenter deployments?

A. SimpleStrategy
B. NetworkTopologyStrategy
C. RandomPartitioner
D. ByteOrderedPartitioner

Answer: B. NetworkTopologyStrategy

31. What is the primary purpose of Cassandra's Bloom Filter?

A. Store key-value pairs permanently
B. Compress SSTables
C. Quickly determine if a key may exist in an SSTable
D. Replace indexing structures

Answer: C. Quickly determine if a key may exist in an SSTable

32. Which statement about Bloom Filters is TRUE?

A. They may produce false negatives.
B. They never produce false positives.
C. They may produce false positives but never false negatives.
D. They guarantee exact membership testing.

Answer: C. They may produce false positives but never false negatives.

33. In Cassandra, why are writes generally faster than reads?

A. Reads use only memory.
B. Writes avoid disk operations entirely.
C. Writes are append-oriented using Commit Logs and MemTables.
D. Reads bypass SSTables.

Answer: C. Writes are append-oriented using Commit Logs and MemTables.

34. Which Cassandra consistency level is the FASTEST for writes?

A. ALL
B. QUORUM
C. ONE
D. ANY

Answer: D. ANY

35. According to CAP theorem, Cassandra primarily chooses:

A. Consistency and Availability
B. Consistency and Partition Tolerance
C. Availability and Partition Tolerance
D. Atomicity and Durability

Answer: C. Availability and Partition Tolerance

36. Which property defines Eventual Consistency?

A. All replicas are instantly synchronized.
B. Replicas converge if updates stop.
C. Reads always return latest value.
D. No stale reads are possible.

Answer: B. Replicas converge if updates stop.

37. What happens if a ZooKeeper ensemble loses majority availability?

A. A new leader is elected immediately.
B. Remaining node becomes leader.
C. No leader can be elected.
D. System switches to standalone mode.

Answer: C. No leader can be elected.

38. Which ZooKeeper design goal ensures that all updates are processed in a globally agreed order?

A. Ordered
B. Replicated
C. Fast
D. Atomic

Answer: A. Ordered

39. What is the default type of znode in ZooKeeper?

A. Ephemeral
B. Sequential
C. Temporary
D. Persistent

Answer: D. Persistent

Apache Spark and RDDs

40. Which statement best explains Spark's fault tolerance mechanism?

A. Every RDD is replicated across all nodes.
B. Every transformation result is written to HDFS.
C. Spark tracks lineage and recomputes lost partitions.
D. Spark checkpoints every operation by default.

Answer: C. Spark tracks lineage and recomputes lost partitions.

41. Which operation below is an Action rather than a Transformation?

A. collect()
B. join()
C. map()
D. filter()

Answer: A. collect()

42. Which Spark component is responsible for splitting a DAG into stages?

A. SparkContext
B. TaskScheduler
C. DAGScheduler
D. BlockManager

Answer: C. DAGScheduler

43. Why is reduceByKey() generally preferred over groupByKey() for aggregation?

A. It automatically sorts keys.
B. It uses combiners on the map side.
C. It creates fewer partitions.
D. It avoids shuffling completely.

Answer: B. It uses combiners on the map side.

44. In Spark, which operation triggers actual execution of a DAG?

A. map()
B. filter()
C. join()
D. count()

Answer: D. count()

45. Which statement about RDDs is FALSE?

A. RDDs are immutable.
B. RDDs can be cached for reuse.
C. RDDs automatically rebuild lost data using lineage.
D. RDDs support direct modification of existing partitions.

Answer: D. RDDs support direct modification of existing partitions.

46. Which feature helps Spark recover lost RDD partitions?

A. Replication
B. Lineage
C. Compression
D. Scheduling

Answer: B. Lineage

47. SparkContext is:

A. Storage system
B. Main entry point to Spark functionality
C. Worker node
D. Scheduler

Answer: B. Main entry point to Spark functionality

48. Which Spark component launches tasks on worker nodes and retries failed tasks?

A. SparkContext
B. DAGScheduler
C. BlockManager
D. TaskScheduler

Answer: D. TaskScheduler

49. A Spark application performs multiple iterations over the same dataset. Which feature provides the greatest performance improvement?

A. HDFS replication
B. Data compression
C. Caching/Persistence of RDDs
D. Increasing block size

Answer: C. Caching/Persistence of RDDs

Ensemble Learning (Random Forest and Boosting)

50. Random Forest is best described as:

A. A single deep decision tree
B. A clustering algorithm
C. A neural network ensemble
D. A bagging of de-correlated decision trees

Answer: D. A bagging of de-correlated decision trees

51. What is the main purpose of Bagging?

A. Increase bias
B. Reduce prediction variance
C. Reduce dataset size
D. Increase overfitting

Answer: B. Reduce prediction variance

52. Bootstrap in Bagging refers to:

A. Sampling with replacement
B. Feature scaling
C. Gradient computation
D. Data normalization

Answer: A. Sampling with replacement

53. In Random Forest, at each split:

A. All features are always evaluated
B. Random subset of features is selected
C. Features are ignored
D. Only one feature exists

Answer: B. Random subset of features is selected

54. Random Forest can naturally support parallel processing because:

A. Trees are built independently
B. Trees share memory
C. Trees use one processor
D. Trees require sequential execution

Answer: A. Trees are built independently

55. Boosting combines:

A. Strong classifiers only
B. Weak classifiers into a strong ensemble
C. Clusters into datasets
D. Databases into warehouses

Answer: B. Weak classifiers into a strong ensemble

56. Gradient Boosting builds models:

A. In a single step
B. Iteratively
C. Randomly
D. Without training

Answer: B. Iteratively

57. In Gradient Boosted Trees, the base learners are usually:

A. Decision Trees
B. Neural Networks
C. K-Means Clusters
D. Support Vectors

Answer: A. Decision Trees

58. Cross-validation is mainly used to:

A. Increase dataset size
B. Assess model quality and select best parameters
C. Reduce number of features
D. Perform clustering

Answer: B. Assess model quality and select best parameters

59. Recommended learning rate for Gradient Boosting is:

A. Greater than 1
B. Equal to 5
C. Less than or equal to 0.1
D. Equal to 10

Answer: C. Less than or equal to 0.1

Spark Streaming

60. What is the fundamental processing model used by Spark Streaming?

A. Record-at-a-time processing
B. Micro-batch processing
C. Batch-only processing
D. Pipeline processing

Answer: B. Micro-batch processing

61. Which abstraction represents a stream of data in Spark Streaming?

A. DataFrame
B. RDD
C. DStream
D. Dataset

Answer: C. DStream

62. Which of the following is NOT a supported Spark Streaming input source?

A. Kafka
B. Flume
C. Twitter
D. HDFS Block Replicator

Answer: D. HDFS Block Replicator

63. Why is maintaining two separate stacks for batch and streaming considered problematic?

A. Reduces throughput
B. Doubles implementation and operational effort
C. Prevents fault tolerance
D. Eliminates scalability

Answer: B. Doubles implementation and operational effort

64. What is the main advantage of Spark Streaming's micro-batch architecture?

A. Eliminates RDDs
B. Supports only offline processing
C. Provides fault tolerance and scalability
D. Requires no cluster

Answer: C. Provides fault tolerance and scalability

65. Which Spark Streaming feature helps recover lost data after worker failure?

A. Replicated input data and RDD lineage
B. ZooKeeper election
C. Kafka partitions
D. SQL optimizer

Answer: A. Replicated input data and RDD lineage

66. Which function maintains arbitrary state across batches?

A. updateStateByKey()
B. reduceByKey()
C. union()
D. collect()

Answer: A. updateStateByKey()

67. In Spark Streaming, every interval results in:

A. A new ZooKeeper node
B. An RDD graph computation
C. A new Kafka broker
D. A new DataNode

Answer: B. An RDD graph computation

68. Which statement about Spark Streaming fault tolerance is TRUE?

A. Lost data cannot be recovered.
B. Transformations are exactly-once and fault-tolerant
C. Only input data is fault tolerant.
D. Failures require manual data recreation.

Answer: B. Transformations are exactly-once and fault-tolerant

69. Which requirement is NOT listed for a stream-processing framework in Spark Streaming?

A. Second-scale latency
B. Efficient fault tolerance
C. Scalability
D. Fixed single-node execution

Answer: D. Fixed single-node execution

Cluster Analysis and K-Means

70. Cluster analysis is primarily a:

A. Supervised learning technique
B. Reinforcement learning technique
C. Unsupervised learning technique
D. Deep learning technique

Answer: C. Unsupervised learning technique

71. What is the main goal of cluster analysis?

A. Predict future values
B. Organize similar items into groups
C. Classify labeled data
D. Reduce storage space

Answer: B. Organize similar items into groups

72. Which similarity measure calculates straight-line distance between two points?

A. Manhattan Distance
B. Cosine Similarity
C. Euclidean Distance
D. Hamming Distance

Answer: C. Euclidean Distance

73. Which similarity measure is based on the angle between vectors?

A. Euclidean Distance
B. Manhattan Distance
C. Cosine Similarity
D. Jaccard Distance

Answer: C. Cosine Similarity

74. Which is a common application of cluster analysis?

A. Customer segmentation
B. Sorting algorithms
C. Compiler optimization
D. CPU scheduling

Answer: A. Customer segmentation

75. Which algorithm is most commonly associated with cluster analysis?

A. Decision Tree
B. Naive Bayes
C. K-Means
D. Logistic Regression

Answer: C. K-Means

76. What is the first step of the K-Means algorithm?

A. Compute accuracy
B. Select initial centroids
C. Normalize output
D. Remove outliers

Answer: B. Select initial centroids

77. Which metric is commonly used to evaluate K-Means clustering quality?

A. Accuracy
B. Recall
C. WSS
D. Precision

Answer: C. WSS

78. If WSS decreases, it generally indicates:

A. Better cluster compactness
B. Worse clustering
C. More classification errors
D. Data loss

Answer: A. Better cluster compactness

79. K-Means stops iterating when:

A. No changes occur in centroids
B. Accuracy becomes 100%
C. Dataset becomes empty
D. Number of features decreases

Answer: A. No changes occur in centroids

GraphX

80. What is the main goal of GraphX?

A. Database management
B. Unify data-parallel and graph-parallel analytics
C. Replace Spark SQL
D. Replace HDFS

Answer: B. Unify data-parallel and graph-parallel analytics

81. Which phase is NOT part of the Graph-Parallel Pattern?

A. Gather
B. Apply
C. Scatter
D. Shuffle Sort

Answer: D. Shuffle Sort

82. Which GraphX operator joins vertices and edges together?

A. mapV()
B. subgraph()
C. triplets
D. reverse()

Answer: C. triplets

83. In GraphX, a Property Graph consists of:

A. Only vertices
B. Only edges
C. Vertices and edges with properties
D. Tables only

Answer: C. Vertices and edges with properties

84. Which graph algorithm is commonly used to identify influential vertices?

A. K-Means
B. PageRank
C. Apriori
D. Frequent Pattern Growth

Answer: B. PageRank

85. Triangle Counting is mainly used for:

A. Sorting vertices
B. Community detection
C. Data compression
D. Replication

Answer: B. Community detection

86. Which optimization reduces communication by avoiding unnecessary joins?

A. Join Elimination
B. Data Duplication
C. Replication Factor
D. Hash Partitioning

Answer: A. Join Elimination

87. Which GraphX feature helps accelerate joins and construct subgraphs efficiently?

A. Bloom Filters
B. Indexing and Bitmaps
C. Commit Logs
D. MemTables

Answer: B. Indexing and Bitmaps

88. Which statement best describes GraphX?

A. Graph-only processing framework
B. Batch-only processing framework
C. Unified analytics framework for graph and data processing
D. SQL replacement system

Answer: C. Unified analytics framework for graph and data processing

89. Which GraphX optimization reuses routing tables, edge adjacency information, and indices across graphs and subgraphs?

A. Substantial Index and Data Reuse
B. Active Edge Indexing
C. Join Elimination
D. Pregel Messaging

Answer: A. Substantial Index and Data Reuse